Organic Cume, April 29, 2016

1. Draw the chair conformation of cyclohexane including all hydrogen positions.

2. Give the energy diagram, with accurately drawn structures and energies, for the chair-chair interconversion of cyclohexane. On this diagram, draw each of the conformations for each step.

X	ΔG^o, kcal/mol (kJ/mol)	X	ΔG^o, kcal/mol (kJ/mol)
H—	0 (0)	Cl—	0.52 (2.2)
CH_3—	1.74 (7.28)	Br—	0.57 (2.4)
CH_3CH_2—	~1.75 (~7.3)	I—	0.45 (1.9)
$(CH_3)_2HC$—	~2.15 (~9.0)	HO—	0.93 (3.9)
$(CH_3)_3C$—	4.9 (21)	H_2N—	1.41 (5.9)
C_6H_{11}—	2.15 (9.0)	CH_3O—	0.75 (3.1)
$HOC(=O)$—	1.41 (5.90)	C_6H_5—	2.7 (11)
		$H_2C{=}CH$—	1.7 (7.1)
$CH_3OC(=O)$—	1.29 (5.40)	$HC{\equiv}C$—	0.41 (1.7)

3. Given the A values in the table above, calculate the energy difference from the equatorial chair to the axial chair conformations in methylcyclohexane.

4. Why is iodocyclohexane's value smaller than that of chloro- or bromocyclohexane?

5. Calculate the energy difference between the two chair conformations in *trans*-1,4-dimethylcyclohexane.

6. Do you expect the energy difference between the two chair conformations in *trans*-1,4-dimethylcyclohexane to be very similar, or not, to that for the two chair conformations in *cis*-1,3,-dimethylcyclohexane? Why?

7. Discuss the reasons for strain energy in cyclopropane. Be clear with appropriate structural drawings and valence bond orbital descriptions.

8. Discuss the reasons for strain energy in cyclobutane. Be clear with appropriate structural drawings and valence bond orbital descriptions.